

Module - I

Complex Analysis I

Function of Complex Variable

If x & y are real variables, then $x + iy$ is called a complex variable.

If corresponding to each value of a complex variable $z = x + iy$ in a given region, there correspond one or more values of another complex variable $w = u + iv$, then w is called a function of the complex variable z , and is denoted by,

$$w = f(z) = u + iv$$

where, u is the real part and v is the imaginary part.

Example:

$$\text{Let, } w = f(z) = z^2$$

$$f(z) = (x + iy)^2 = x^2 + (iy)^2 + 2ixy = x^2 - y^2 + 2ixy$$

Comparing with $f(z) = u + iv$,

$$u + iv = (x^2 - y^2) + i(2xy)$$

Here, u & v are functions of the real variables x & y .

Derivative of $f(z)$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} \quad [\Delta z \text{ or } \delta z]$$

Differentiability of $f(z)$

If the derivative of $f(z)$ is unique or independent of the manner in which $\delta z \rightarrow 0$, then the function is called differentiable at that point.

Analytic Function

A function $f(z)$ is said to be analytic at a point $z = z_0$ if f is differentiable not only at z_0 , but also at every point in some neighbourhood of z_0 .

Example:

$$f(z) = e^z, \sin z, \cos z, \text{etc.}$$

A function is analytic in a domain if it is analytic at every point of that domain.

An analytic function is also known as a holomorphic, regular or monogenic function.

Entire Function

A function which is analytic everywhere (for all set of points in the complex plane) is known as an Entire Function.

Example (Polynomial):

$$f(z) = a_0 + a_1z + a_2z^2 + \dots + a_nz^n$$

Entire Function $\Rightarrow$ Analytic $\Rightarrow$ Differentiable $\Rightarrow$ Continuous

Cauchy - Riemann Equations (C-R Equations)

It is used to determine whether a complex function is analytic, provides necessary and sufficient conditions.

Necessary Condition for $f(z)$ to be Analytic:

The necessary condition for a function $f(z) = u + iv$ to be analytic at all points in a region R is,

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ ---- (1) \quad \& \quad } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ ---- (2)}$$

provided that the partial derivatives exist.

Equations (1) & (2) are called the Cauchy-Riemann Equations.

Sufficient Condition for $f(z)$ to be Analytic:

For $f(z) = u + iv$ to be analytic in a region R ,

1. $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ & $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

2. The partial derivatives $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ are continuous functions of x & y in region R .

Q. Find the values of c_1 & c_2 such that the function,

$f(z) = (x^2 + c_1 y^2 - 2xy) + i(c_2 x^2 - y^2 + 2xy)$ is analytic. Also find $f'(z)$.

Solⁿ:

$$\text{Here, } u = x^2 + c_1 y^2 - 2xy \\ \& \quad v = c_2 x^2 - y^2 + 2xy$$

Since, $f(z)$ is analytic, it satisfies the C-R equations.

$$\text{i.e. } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ --- (1) \& } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ --- (2)}$$

Now,

$$\frac{\partial u}{\partial x} = 2x - 2y$$

$$\frac{\partial v}{\partial y} = -2y + 2x$$

$$\frac{\partial u}{\partial y} = 2c_1 y - 2x$$

$$\frac{\partial v}{\partial x} = 2c_2 x + 2y$$

From eqⁿ. (2),

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow 2c_1 y - 2x = -(2c_2 x + 2y)$$

$$\Rightarrow 2c_1 y - 2x = -2c_2 x - 2y$$

Comparing coefficients of x & y , we get,

$$2c_1 = -2 \Rightarrow c_1 = -1.$$

$$\& -2 = -2c_2 \Rightarrow c_2 = 1.$$

We know,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = (2x - 2y) + i(2x + 2y)$$

$$\Rightarrow f'(z) = 2x - 2y + i2x + i2y$$

$$\Rightarrow f'(z) = 2(x + iy) + 2i\left(x - \frac{1}{i}y\right)$$

$$\Rightarrow f'(z) = 2(x + iy) + 2i(x + iy) \quad \left[\frac{1}{i} = -i\right]$$

$$\Rightarrow f'(z) = 2z + 2iz = 2z(1 + i).$$

Q. Find the constant values of a, b & c such that the function $f(z) = (-x^2 + xy + y^2) + i(ax^2 + bxy + cy^2)$ is analytic. Express $f(z)$ in terms of z .

Solⁿ:

$$\text{Here, } u = -x^2 + xy + y^2 \\ \& \ v = ax^2 + bxy + cy^2$$

Since, $f(z)$ is analytic, it satisfies the C-R eqⁿ.

$$\text{i.e. } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ — (1)}$$

$$\& \ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ — (2)}$$

Now,

$$\frac{\partial u}{\partial x} = -2x + y$$

$$\frac{\partial u}{\partial y} = x + 2y$$

$$\frac{\partial v}{\partial x} = 2ax + by$$

$$\frac{\partial v}{\partial y} = bx + 2cy$$

From eqⁿ. (1),

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\Rightarrow -2x + y = bx + 2cy$$

$$\Rightarrow b = -2 \text{ (On comparing coeff. of } x)$$

$$\& c = 1/2 \text{ (On comparing coeff. of } y)$$

From eqⁿ. (2),

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow x + 2y = -(2ax + by)$$

$$\Rightarrow x + 2y = -2ax - by$$

$$\Rightarrow 1 = -2a \text{ (On comparing coeff. of } x)$$

$$\Rightarrow a = -1/2$$

$$\therefore a = -\frac{1}{2}, b = -2 \text{ \& } c = \frac{1}{2}$$

Then, we have,

$$f(z) = u + iv$$

$$\Rightarrow f(z) = (-x^2 + xy + y^2) + i(ax^2 + bxy + cy^2)$$

$$\Rightarrow f(z) = (-x^2 + xy + y^2) + i \left(\frac{-x^2}{2} - 2xy + \frac{y^2}{2} \right)$$

We know,

$$z = x + iy$$

$$\Rightarrow z^2 = (x + iy)^2$$

$$\Rightarrow z^2 = x^2 + 2ixy + (iy)^2$$

$$\Rightarrow z^2 = x^2 - y^2 + 2ixy$$

$$f(z) = -x^2 + xy + y^2 - \frac{ix^2}{2} - 2ixy + \frac{iy^2}{2}$$

$$\Rightarrow f(z) = -(x^2 - y^2 + 2ixy) - \frac{i}{2} \left(x^2 - y^2 - \frac{2}{i} xy \right)$$

$$\Rightarrow f(z) = -(x^2 - y^2 + 2ixy) - \frac{i}{2} (x^2 - y^2 + 2ixy)$$

$$\Rightarrow f(z) = -z^2 - \frac{iz^2}{2}$$

$$\Rightarrow f(z) = -\frac{z^2}{2} (2+i).$$

Q. Prove that $\sinh(z)$ is analytic. Find its derivatives.

Solⁿ:

$$f(z) = \sinh z = \sinh(x+iy)$$

$$\Rightarrow f(z) = \sinh(x) \cdot \cosh(iy) + \sinh(iy) \cdot \cosh(x)$$

$$[\sinh(a+b) = \sinh(a) \cdot \cosh(b) + \sinh(b) \cdot \cosh(a)]$$

$$\Rightarrow f(z) = u + iv$$

We know,

$$(1) \sin x = \frac{e^{ix} - e^{-ix}}{2i}$$

$$(2) \cos x = \frac{e^{ix} + e^{-ix}}{2}$$

$$(3) \sinh x = \frac{e^x - e^{-x}}{2}$$

$$(4) \cosh x = \frac{e^x + e^{-x}}{2}$$

$$\Rightarrow \cosh(iy) = \frac{e^{iy} + e^{-iy}}{2} = \cos y$$

$$\& \sinh(iy) = \frac{e^{iy} - e^{-iy}}{2} = i \left(\frac{e^{iy} - e^{-iy}}{2i} \right) = i \sin y$$

$$\text{So, } f(z) = (\sinh x \cdot \cos y) + i (\sin y \cdot \cosh x)$$

$$\text{Here, } u = \sinh x \cdot \cos y \\ \& \ v = \sin y \cdot \cosh x$$

Now,

$$\frac{\partial u}{\partial x} = \cosh x \cdot \cos y$$

$$\frac{\partial u}{\partial y} = -\sinh x \cdot \sin y$$

$$\frac{\partial v}{\partial x} = \sin y \cdot \sinh x$$

$$\frac{\partial v}{\partial y} = \cos y \cdot \cosh x$$

$$\text{Here, } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore$ C-R eqⁿ. are satisfied.

$\because \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}$ & $\frac{\partial v}{\partial y}$ are continuous
function of x, y and $f(z)$ satisfies C-R eqⁿ.

$\therefore f(z)$ is analytic.

For derivative of $f(z)$,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow f'(z) = (\cosh x \cdot \cos y) + i(\sinh x \cdot \sin y)$$

$$\Rightarrow f'(z) = \cosh x \cdot \cosh(iy) + \sinh x \cdot \sinh(iy)$$

$$\Rightarrow f'(z) = \cosh(x+iy) \quad [\cosh(x+iy) = \cosh x \cdot \cosh y + \sinh x \cdot \sinh y]$$

$$\Rightarrow f'(z) = \cosh(z).$$

9. Show that if $f(z)$ is analytic and real part of $f(z)$ is constant then $f(z)$ is constant.

Solⁿ:

We know,

$$f(z) = u + iv \quad (\text{Given } u \text{ is constant})$$

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\text{Since, } u \text{ is constant, } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = 0.$$

Also, $f(z)$ is analytic, it satisfies C-R eqⁿ.

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow \frac{\partial v}{\partial x} = 0$$

$$\Rightarrow f'(z) = 0$$

Thus, $f(z)$ is constant.

Q. Verify if $f(z) = \frac{xy^2(x+iy)}{x^2+y^4}$, $z \neq 0$, $f(0) = 0$ is analytic or not.

Solⁿ:

$$f(z) = \frac{xy^2(x+iy)}{x^2+y^4}$$

$$\Rightarrow f(z) = \left(\frac{x^2y^2}{x^2+y^4} \right) + i \left(\frac{xy^3}{x^2+y^4} \right)$$

Here, $u = \frac{x^2y^2}{x^2+y^4}$ & $v = \frac{xy^3}{x^2+y^4}$

Now,

$$\frac{\partial u}{\partial x} = \lim_{x \rightarrow 0} \frac{u(x, 0) - u(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0-0}{x} = 0$$

$$\frac{\partial u}{\partial y} = \lim_{y \rightarrow 0} \frac{u(0, y) - u(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0-0}{y} = 0$$

$$\frac{\partial v}{\partial x} = \lim_{x \rightarrow 0} \frac{v(x, 0) - v(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0-0}{x} = 0$$

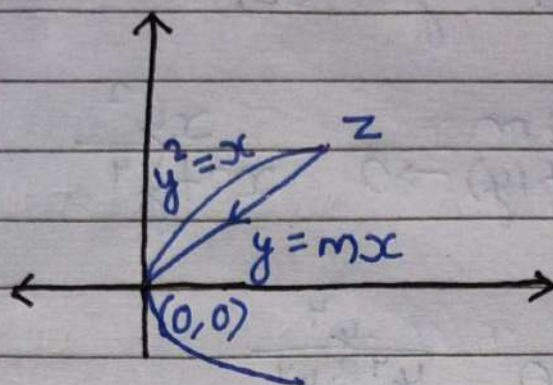
$$\frac{\partial v}{\partial y} = \lim_{y \rightarrow 0} \frac{v(0, y) - v(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0-0}{y} = 0$$

$$\text{i.e. } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad \text{at } z=0$$

and the above partial derivatives are continuous.

Then,

$$f'(z) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z}$$



Let, $z \rightarrow 0$ along the straight line, $y = mx$.

$$f'(z) = \lim_{(x+iy) \rightarrow 0} \frac{\frac{xy^2(x+iy)}{x^2+y^4} - 0}{(x+iy)}$$

$$\Rightarrow f'(z) = \lim_{(x+iy) \rightarrow 0} \frac{xy^2(x+iy)}{(x^2+y^4)(x+iy)}$$

$$\Rightarrow f'(z) = \lim_{(x+iy) \rightarrow 0} \frac{xy^2}{x^2+y^4}$$

$$\Rightarrow f'(z) = \lim_{(x+iy) \rightarrow 0} \frac{xy^2}{x^2+y^4}$$

$$\Rightarrow f'(z) = \lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2 + m^4 x^4} \quad \begin{matrix} [y = mx] \\ [x + imx \rightarrow 0, x \rightarrow 0] \end{matrix}$$

$$\Rightarrow f'(z) = \lim_{x \rightarrow 0} \frac{m^2 x^3}{x^2(1 + m^4 x^2)}$$

$$\Rightarrow f'(z) = \lim_{x \rightarrow 0} \frac{m^2 x}{1 + m^4 x^2} = 0$$

Along parabola $y^2 = x$,

$$f'(z) = \lim_{(x+iy) \rightarrow 0} \frac{xy^2}{x^2 + y^4}$$

$$\Rightarrow f'(z) = \lim_{y \rightarrow 0} \frac{y^4}{y^4 + y^4}$$

$$\Rightarrow f'(z) = \lim_{y \rightarrow 0} \frac{y^4}{2y^4}$$

$$\Rightarrow f'(z) = \frac{1}{2}$$

$\therefore f'(z)$ depends on the path along which $z \rightarrow 0$.

$\therefore f(z)$ is not differentiable and not analytic at $z=0$ for all the angle.

9. If $f(z) = \frac{x^3 y (y - ix)}{x^6 + y^2}$, $z \neq 0$, Prove that:
 0 , $z = 0$ } $\frac{f(z) - f(0)}{z} \rightarrow 0$

as $z \rightarrow 0$ along any radius vector but not as $z \rightarrow 0$ in any manner and also that $f(z)$ is not analytic at $z = 0$.

Solⁿ:

$$f'(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z}$$

Let $z \rightarrow 0$ along radius vector $y = mx$,

$$f'(0) = \lim_{(x+iy) \rightarrow 0} \frac{\cancel{f'(0)} \frac{x^3 y (y - ix)}{x^6 + y^2} - 0}{(x+iy)}$$

$$\Rightarrow f'(0) = \lim_{(x+iy) \rightarrow 0} \frac{x^3 y (y - ix)}{(x^6 + y^2)(x+iy)}$$

$$\Rightarrow f'(0) = \lim_{(x+iy) \rightarrow 0} \frac{-i x^3 y (x+iy)}{(x^6 + y^2)(x+iy)}$$

$$\Rightarrow f'(0) = \lim_{(x+iy) \rightarrow 0} \frac{-i x^3 y}{x^6 + y^2}$$

Along $y = mx$,

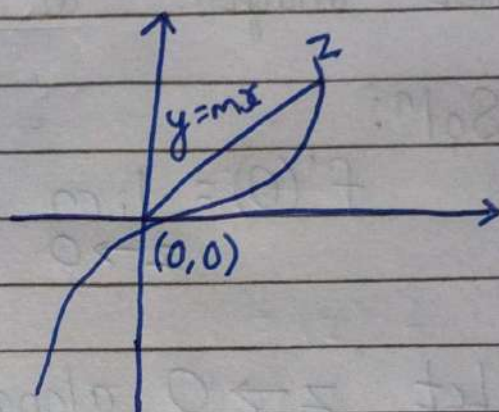
$$f'(0) = \lim_{x \rightarrow 0} \frac{-ix^3(mx)}{x^6 + m^2x^2} \quad \left[\begin{array}{l} x+iy \rightarrow 0 \\ x+imx \rightarrow 0 \Rightarrow x \rightarrow 0 \end{array} \right]$$

$$= \lim_{x \rightarrow 0} \frac{-imx^4}{x^6 + m^2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{-imx^4 \cdot x^{-2}}{x^2(x^4 + m^2)}$$

$$= \lim_{x \rightarrow 0} \frac{-imx^2}{x^4 + m^2}$$

$$= 0.$$



Let $z \rightarrow 0$ along the curve $y = -x^3$,

$$f'(0) = \lim_{\substack{(x) \rightarrow 0 \\ (iy) \rightarrow 0}} \frac{-ix^3y}{x^6 + y^2}$$

$$\Rightarrow f'(0) = \lim_{x \rightarrow 0} \frac{ix^6}{x^6 + x^6} = \lim_{x \rightarrow 0} \frac{ix^6}{2x^6} = \frac{i}{2}.$$

$\therefore f'(0)$ depends on the path along which $z \rightarrow 0$.

$\therefore f'(0)$ does not exist.

Hence, $f(z)$ is not differentiable at $z=0$ and thus not analytic at $z=0$.

Q. Examine the nature of the function,

$$f(z) = \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}} ; z \neq 0 \text{ and}$$

$f(0) = 0$ including at the origin.

Solⁿ:

$$f(z) = \left(\frac{x^3 y^5}{x^4 + y^{10}} \right) + i \left(\frac{x^2 y^6}{x^4 + y^{10}} \right)$$

$$\text{Here, } u = \frac{x^3 y^5}{x^4 + y^{10}} \quad \& \quad v = \frac{x^2 y^6}{x^4 + y^{10}}$$

Now,

$$\frac{\partial u}{\partial x} = \lim_{x \rightarrow 0} \frac{u(x, 0) - u(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0$$

$$\frac{\partial u}{\partial y} = \lim_{y \rightarrow 0} \frac{u(0, y) - u(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0 - 0}{y} = 0$$

$$\frac{\partial v}{\partial x} = \lim_{x \rightarrow 0} \frac{v(x, 0) - v(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0$$

$$\frac{\partial v}{\partial y} = \lim_{y \rightarrow 0} \frac{v(0, y) - v(0, 0)}{y} = \lim_{y \rightarrow 0} \frac{0 - 0}{y} = 0$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$\therefore$ C-R eqⁿ are satisfied at the origin.

$$\text{Now, } f'(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z}$$

$$\Rightarrow f'(0) = \lim_{(x+iy) \rightarrow 0} \frac{\frac{x^2 y^5 (x+iy)}{x^4 + y^{10}} - 0}{x+iy}$$

$$= \lim_{(x+iy) \rightarrow 0} \frac{x^2 y^5 \cancel{(x+iy)}}{(x^4 + y^{10}) \cancel{(x+iy)}}$$

$$= \lim_{(x+iy) \rightarrow 0} \frac{x^2 y^5}{x^4 + y^{10}}$$

Let, $z \rightarrow 0$ along $y=x$,

$$f'(0) = \lim_{x \rightarrow 0} \frac{x^7}{x^4 + x^{10}} = \lim_{x \rightarrow 0} \frac{x^7 \cdot x^3}{x^4 (1 + x^6)} = \lim_{x \rightarrow 0} \frac{x^3}{1 + x^6}$$

$$\Rightarrow f'(0) = 0$$

Let, $z \rightarrow 0$ along $y^5 = x^2$

$$f'(0) = \lim_{x \rightarrow 0} \frac{x^4}{x^4 + x^4} = \lim_{x \rightarrow 0} \frac{x^4}{2x^4} = \frac{1}{2}$$

$\therefore f'(0)$ does not exist.

$\therefore f(z)$ is not differentiable at $z=0$.

So, $f(z)$ is not analytic at origin although it satisfies C-R eqⁿ. at the origin.

Cauchy - Riemann Equation in Polar Coordinate System

$$z = x + iy$$

$$x = r \cos \theta \quad \& \quad y = r \sin \theta$$

$$\Rightarrow z = r \cos \theta + i(r \sin \theta)$$

$$\Rightarrow z = r (\cos \theta + i \sin \theta)$$

$$\Rightarrow z = r e^{i\theta}$$

Also,

$$f(z) = u + iv$$

$$\Rightarrow f(z) = u(r, \theta) + i v(r, \theta)$$

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$$

$$\& \frac{\partial v}{\partial r} = -\frac{1}{r} \frac{\partial u}{\partial \theta}$$

C-R Equation in Polar Coordinate System

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = (\cos \theta - i \sin \theta) \frac{\partial f}{\partial r}$$

$$\& f'(z) = -\frac{i}{r} (\cos \theta - i \sin \theta) \frac{\partial f}{\partial \theta}$$

Q. Find the value of p such that: $f(z) = z^2 \cos 2\theta + i z^2 \sin p\theta$ is analytic.

Solⁿ:

$$\text{Given, } u = z^2 \cos 2\theta \\ v = z^2 \sin p\theta$$

$$\frac{\partial u}{\partial x} = 2x \cos 2\theta$$

$$\frac{\partial v}{\partial x} = 2x \sin p\theta$$

$$\frac{\partial u}{\partial \theta} = -2z^2 \sin 2\theta$$

$$\frac{\partial v}{\partial \theta} = pz^2 \cos p\theta$$

$\therefore f(z)$ is analytic.

$\therefore$ It satisfies C-R equation.

$$\text{i.e. } \frac{\partial u}{\partial x} = \frac{1}{x} \frac{\partial v}{\partial \theta} \quad \& \quad \frac{\partial v}{\partial x} = -\frac{1}{x} \frac{\partial u}{\partial \theta}$$

$$\text{Taking, } \frac{\partial u}{\partial x} = \frac{1}{x} \frac{\partial v}{\partial \theta}$$

$$\Rightarrow 2r \cos 2\theta = \frac{1}{r} (pr^2 \cos p\theta)$$

$$\Rightarrow 2r \cos 2\theta = pr \cos p\theta$$

$$\Rightarrow 2 \cos 2\theta = p \cos p\theta$$

$$\Rightarrow p = 2.$$

Q. Show that $f(z) = \log z$ is analytic everywhere except at the origin and its derivative is $1/z$.

Solⁿ:

$$f(z) = \log z = \log(x+iy) \text{ (Not Possible)}$$

From Polar Coordinate System,
 $z = r.e^{i\theta}$

$$f(z) = \log(r.e^{i\theta})$$

$$\Rightarrow f(z) = \log r + \log(e^{i\theta})$$

$$\Rightarrow f(z) = \log r + i\theta(\log e)$$

$$\Rightarrow f(z) = \log r + i\theta$$

Here, $u = \log r$ & $v = \theta$ [$f(z) = u + iv$]

Now,

$$\frac{\partial u}{\partial x} = \frac{1}{x}$$

$$\frac{\partial u}{\partial \theta} = 0$$

$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial v}{\partial \theta} = 1$$

Here,

$$\frac{\partial u}{\partial x} = \frac{1}{x} \quad \frac{\partial v}{\partial \theta} = \frac{1}{x} \quad \checkmark$$

$$\frac{\partial v}{\partial x} = -\frac{1}{x} \quad \frac{\partial u}{\partial \theta} = 0 \quad \checkmark$$

At the origin, $z = 0$

$$\Rightarrow x = y = 0$$

$$\Rightarrow r = 0$$

$$[r^2 = x^2 + y^2]$$

At the origin, $\frac{\partial u}{\partial x} = \frac{1}{x}$ does not exist.

$\therefore$ C-R eqⁿ. is not satisfied at the origin.

$\therefore f(z)$ is not analytic at $z = 0$.

For $z \neq 0$, C-R eq^{ns}. are satisfied and $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial x}$, $\frac{\partial u}{\partial \theta}$ & $\frac{\partial v}{\partial \theta}$ are continuous.

So, $f(z)$ is analytic except at the origin.

$$f'(z) = (\cos \theta - i \sin \theta) \frac{\partial f}{\partial h}$$

$$\Rightarrow f'(z) = e^{-i\theta} \frac{1}{x} = \frac{1}{x e^{i\theta}}$$

$$\Rightarrow f'(z) = \frac{1}{z} \quad \underline{\text{proved}}$$

Q. Prove that an analytic function with constant modulus is constant.

Solⁿ:

$$\text{Let } f(z) = u + iv$$

$$\Rightarrow |f(z)| = \sqrt{u^2 + v^2}$$

$$\Rightarrow |f(z)|^2 = u^2 + v^2 = c \text{ (constant)} \text{ --- (1)}$$

On partially differentiating eqⁿ. (1) w.r.to x ,

$$2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} = 0$$

$$\Rightarrow u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} = 0 \text{ --- (2)}$$

Similarly on partially diff. eqⁿ. (1) w.r.to y ,

$$2u \frac{\partial u}{\partial y} + 2v \frac{\partial v}{\partial y} = 0$$

$$\Rightarrow u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y} = 0 \text{ --- (3)}$$

$\therefore f(z)$ is analytic.

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

From eqⁿ. (3),

$$-u \frac{\partial v}{\partial x} + v \frac{\partial u}{\partial x} = 0 \quad (4)$$

On squaring eqⁿ. (2) & (4) and adding,

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\begin{aligned} \Rightarrow |f'(z)|^2 &= \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + u^2 \left(\frac{\partial u}{\partial x}\right)^2 + v^2 \left(\frac{\partial v}{\partial x}\right)^2 \\ &\quad + 2uv \left(\frac{\partial u}{\partial x}\right) \left(\frac{\partial v}{\partial x}\right) + u^2 \left(\frac{\partial v}{\partial x}\right)^2 + v^2 \left(\frac{\partial u}{\partial x}\right)^2 \\ &\quad - 2uv \left(\frac{\partial u}{\partial x}\right) \left(\frac{\partial v}{\partial x}\right) = 0 \end{aligned}$$

$$\Rightarrow u^2 \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right] + v^2 \left[\left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial x}\right)^2 \right] = 0$$

$$\Rightarrow (u^2 + v^2) \left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \right] = 0$$

$$\Rightarrow \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 = 0$$

$$\Rightarrow |f'(z)|^2 = 0$$

$$\Rightarrow |f'(z)| = 0 \Rightarrow \frac{\partial u}{\partial x} = \frac{\partial v}{\partial x} = 0$$

$$\Rightarrow u = v = \text{constant}$$

$$\Rightarrow f(z) = u + iv = \text{constant.}$$

Q. If $f(z)$ is regular function of z , prove that:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = 4 |f'(z)|^2$$

Solⁿ:

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2$$

$$= \frac{\partial^2}{\partial x^2} |f(z)|^2 + \frac{\partial^2}{\partial y^2} |f(z)|^2$$

We have,

$$f(z) = u + iv$$

$$|f(z)|^2 = u^2 + v^2 = \phi$$

$$\Rightarrow \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) |f(z)|^2 = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$$

Now,

$$\frac{\partial \phi}{\partial x} = 2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x}$$

$$\frac{\partial^2 \phi}{\partial x^2} = 2u \frac{\partial^2 u}{\partial x^2} + 2 \left(\frac{\partial u}{\partial x} \right)^2 + 2v \frac{\partial^2 v}{\partial x^2} + 2 \left(\frac{\partial v}{\partial x} \right)^2 \quad \text{--- (1)}$$

Similarly,

$$\frac{\partial^2 \phi}{\partial y^2} = 2u \frac{\partial^2 u}{\partial y^2} + 2 \left(\frac{\partial u}{\partial y} \right)^2 + 2v \frac{\partial^2 v}{\partial y^2} + 2 \left(\frac{\partial v}{\partial y} \right)^2 \quad (2)$$

Then,

$$\begin{aligned} \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} &= 2u \frac{\partial^2 u}{\partial x^2} + 2 \left(\frac{\partial u}{\partial x} \right)^2 + 2v \frac{\partial^2 v}{\partial x^2} \\ &\quad + 2 \left(\frac{\partial v}{\partial x} \right)^2 + 2u \frac{\partial^2 u}{\partial y^2} + 2 \left(\frac{\partial u}{\partial y} \right)^2 \\ &\quad + 2v \frac{\partial^2 v}{\partial y^2} + 2 \left(\frac{\partial v}{\partial y} \right)^2 \end{aligned}$$

$\therefore f(z)$ is a regular function.

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y}$$

$$\frac{\partial^2 v}{\partial x^2} = \frac{-\partial^2 u}{\partial x \partial y}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{-\partial^2 v}{\partial x \partial y}$$

$$\frac{\partial^2 v}{\partial y^2} = \frac{\partial^2 u}{\partial x \partial y}$$

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 2u \left(\frac{\partial^2 v}{\partial x \partial y} \right) + 2 \left(\frac{\partial u}{\partial x} \right)^2 - 2v \left(\frac{\partial^2 u}{\partial x \partial y} \right)$$

$$+ 2 \left(\frac{\partial v}{\partial x} \right)^2 - 2u \left(\frac{\partial^2 v}{\partial x \partial y} \right) + 2 \left(\frac{\partial v}{\partial x} \right)^2$$

$$+ 2v \left(\frac{\partial^2 u}{\partial x \partial y} \right) + 2 \left(\frac{\partial u}{\partial x} \right)^2$$

$$= 4 \left(\frac{\partial u}{\partial x} \right)^2 + 4 \left(\frac{\partial v}{\partial x} \right)^2$$

$$= 4 \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right]$$

$$= 4 |f'(z)|^2 \quad \underline{\text{proved.}}$$

9. If $f(z)$ is analytic, prove that:

$$\left[\frac{\partial}{\partial x} |f| \right]^2 + \left[\frac{\partial}{\partial y} |f| \right]^2 = |f'|^2$$

Solⁿ:

$$f(z) = u + iv$$

$$|f| = \sqrt{u^2 + v^2} = (u^2 + v^2)^{1/2} \Rightarrow |f|^2 = u^2 + v^2$$

$$\frac{\partial}{\partial x} |f| = \frac{2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x}}{2\sqrt{u^2 + v^2}} = \frac{u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x}}{|f|}$$

Similarly,

$$\frac{\partial}{\partial y} |f| = \frac{u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y}}{|f|}$$

Squaring both sides and adding them,

$$\left[\frac{\partial}{\partial x} |f| \right]^2 + \left[\frac{\partial}{\partial y} |f| \right]^2 = \frac{(u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x})^2}{|f|^2} + \frac{(u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y})^2}{|f|^2}$$

$$\left[u^2 \left(\frac{\partial u}{\partial x} \right)^2 + v^2 \left(\frac{\partial v}{\partial x} \right)^2 + 2uv \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} \right]$$

$$= \frac{+ \left[u^2 \left(\frac{\partial u}{\partial y} \right)^2 + v^2 \left(\frac{\partial v}{\partial y} \right)^2 + 2uv \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right]}{|f|^2}$$

$\therefore f(z)$ is analytic.

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\text{So, } u^2 \left(\frac{\partial u}{\partial x} \right)^2 + v^2 \left(\frac{\partial v}{\partial x} \right)^2 + \cancel{2uv \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x}} + u^2 \left(\frac{\partial v}{\partial x} \right)^2 \\ + v^2 \left(\frac{\partial u}{\partial x} \right)^2 - \cancel{2uv \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x}} \\ \hline |f|^2$$

$$= \frac{(u^2 + v^2) \left(\frac{\partial u}{\partial x} \right)^2 + (u^2 + v^2) \left(\frac{\partial v}{\partial x} \right)^2}{|f|^2}$$

$$= \frac{(u^2 + v^2) \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right]}{|f|^2}$$

$$= \frac{\cancel{(u^2 + v^2)} \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \right]}{\cancel{(u^2 + v^2)}}$$

$$\Rightarrow \left[\frac{\partial}{\partial x} |f| \right]^2 + \left[\frac{\partial}{\partial y} |f| \right]^2 = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 \quad \text{--- (1)}$$

We know,
 ~~$f'(z)$~~

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\Rightarrow |f'(z)|^2 = \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 \quad \text{--- (2)}$$

From eqⁿ. (1) & (2),

$$\left[\frac{\partial}{\partial x} |f|\right]^2 + \left[\frac{\partial}{\partial y} |f|\right]^2 = |f'(z)|^2 \quad \text{proved}$$